

CC-215 Database Systems
ASSIGNMENT#1
BSSE - Fall 2024 – Semester Spring 2025

Total Marks: 100

Deadline: April 10, 2025

Instructions

For each question, you must perform the following tasks:

- a) List out entity classes along with appropriate attributes.
- b) Identify all necessary relationships.
- c) Identify the connectivity for each relationship.
- d) Draw ER / EER model using an appropriate CASE tool, such as MS Visio, Lucid chart etc.
- e) You can make your assumptions (if needed).

Question Set – 1 [4 Questions, 10 Marks Each]

Question#1:

In the PDO Company, we have Identified the following entity type. An Identified is also suggested for each entity together with selected important attributes. The company receive many orders from different customers. The identifier for an Employee is EMPLOYED the following additional attributes are Employee Name, Employee Number, Employee birth date, Age which will get based on the Birthday and Employee qualification (some employees are having more than One qualification and we have other Entities In these company Sales, STOCK and INVENTORY for Employees who are working here and The FUEL entity have Flowing attributes are Price of Fuel and Number of Tanks. An employee delivers any number of orders each order must belong to exactly one employee custom MER Submit order. The identity of ORDER is order-Number Other attributes include order- Name and order-Date.

Services:

- Customer can place order
- Order taking Employee will receive customer request.
- Employee will check the inventory and proceed the customer request.
- Employees have their designations according to their qualifications in the company.
- Delivery employee receive the fuel tanks from company to deliver the order to customer

Question#2:

Prepare an ERD for a real estate firm that lists property for sale. Also prepare a definition for each entity type, attribute, and relationship on your diagram. The following describes this organization:

- The firm has a number of sales offices in several states. Attributes of sales office include Office Number (identifier) and Location.
- Each sales office is assigned one or more employees. Attributes of employee include Employee ID (identifier) and Employee Name. An employee must be assigned to only one sales office.
- For each sales office, there is always one employee assigned to manage that office. An employee may manage only the sales office to which he or she is assigned.
- The firm lists property for sale. Attributes of property include Property ID (identifier) and Location. Components of Location include Address, City, State, and Zip Code.
- Each unit of property must be listed with one (and only one) of the sales offices. A sales office may have any number of properties listed or may have no properties listed.
- Each unit of property has one or more owners. Attributes of owners are Owner ID (identifier) and Owner Name. An owner may own one or more units of property. An attribute of the relationship between property and owner is Percent Owned.

Question#3:

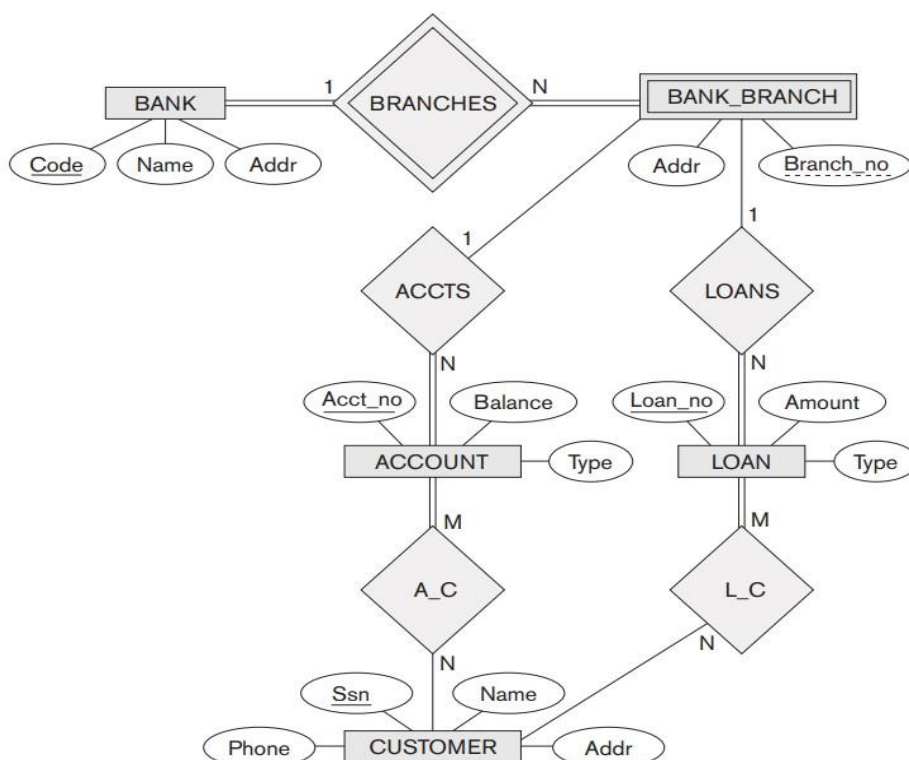
Create an ERD for a car dealership. The dealership sells both new and used cars, and it operates a service facility. Base your design on the following business rules:

- A salesperson may sell many cars, but each car is sold by only one salesperson.
- A customer may buy many cars, but each car is bought by only one customer.
- A salesperson writes a single invoice for each car he or she sells.
- A customer gets an invoice for each car he or she buys.
- A customer may come in just to have his or her car serviced; that is, a customer need not buy a car to be classified as a customer.
- When a customer takes one or more cars in for repair or service, one service ticket is written for each car.
- The car dealership maintains a service history for each of the cars serviced. The service records are referenced by the car's serial number.
- A car brought in for service can be worked on by many mechanics, and each mechanic may work on many cars.
- A car that is serviced may or may not need parts (e.g., adjusting a carburetor or cleaning a fuel injector nozzle does not require providing new parts).

Question#4:

Consider the ER diagram shown in the following figure for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

- List the strong entity types in the ER diagram.
- Is there a weak entity type? If so, give its name, partial key, and identifying relationship.
- What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?
- List the names of all relationships.
- List concisely the user requirements that led to this ER schema design.



Question Set – 2 [4 Questions, 15 Marks Each]

Question# 1:

During peak periods, Temporary Employment Corporation (TEC) places temporary workers in companies. TEC's manager gives you the following description of the business:

- TEC has a file of candidates who are willing to work.
- Any candidate who has worked before has a specific job history. (Naturally, no job history exists if the candidate has never worked.) Each time the candidate works, one additional job history record is created.
- Each candidate has earned several qualifications. Each qualification may be earned by more than one candidate. (For example, more than one candidate may have earned a Bachelor of Business Administration degree or a Microsoft Network Certification, and clearly a candidate may have earned both a BBA and a Microsoft Network Certification.)
- TEC offers courses to help candidates improve their qualifications.
- Every course develops one specific qualification; however, TEC does not offer a course for every qualification. Some qualifications are developed through multiple courses.
- Some courses cover advanced topics that require specific qualifications as prerequisites. Some courses cover basic topics that do not require any prerequisite qualifications. A course can have several prerequisites. A qualification can be a prerequisite for more than one course.
- Courses are taught during training sessions. A training session is the presentation of a single course. Over time, TEC will offer many training sessions for each course; however, new courses may not have any training sessions scheduled right away.
- Candidates can pay a fee to attend a training session. A training session can accommodate several candidates, although new training sessions will not have any candidates registered at first.
- TEC also has a list of companies that request temporaries.
- Each time a company requests a temporary employee, TEC makes an entry in the Openings folder. That folder contains an opening number, a company name, required qualifications, a starting date, an anticipated ending date, and hourly pay.
- Each opening requires only one specific or main qualification.
- When a candidate matches the qualification, the job is assigned, and an entry is made in the Placement Record folder. The folder contains such information as an opening number, candidate number, and total hours worked. In addition, an entry is made in the job history for the candidate.
- An opening can be filled by many candidates, and a candidate can fill many openings.
- TEC uses special codes to describe a candidate's qualifications for an opening. The list of Codes are:

CODE	DESCRIPTION
SEC-45	Secretarial work; candidate must type at least 45 words per minute
SEC-60	Secretarial work; candidate must type at least 60 words per minute
CLERK	General clerking work
PRG-VB	Programmer, Visual Basic
PRG-C++	Programmer, C++
DBA-ORA	Database Administrator, Oracle
DBA-DB2	Database Administrator, IBM DB2
DBA-SQLSERV	Database Administrator, MS SQL Server
SYS-1	Systems Analyst, level 1
SYS-2	Systems Analyst, level 2
NW-NOV	Network Administrator, Novell experience
WD-CF	Web Developer, ColdFusion

Question# 2:

An international school of technology has hired you to create a database management system to assist in scheduling classes. After several interviews with the president, you have come up with the following list of entities, attributes, and initial business rules:

- Room is identified by Building ID and Room No and also has a Capacity. A room can be either a lab or a classroom. If it is a classroom, it has an additional attribute called Board Type.
- Media is identified by MType ID and has attributes of Media Type and Type Description. Note: Here we are tracking type of media (such as a VCR, projector, etc.), not the individual piece of equipment. Tracking of equipment is outside the scope of this project.
- Computer is identified by CType ID and has attributes Computer Type, Type Description, Disk Capacity, and Processor Speed. Please note: As with Media Type, we are tracking only the type of computer, not an individual computer. You can think of this as a class of computers (e.g., PIII 900MHZ).
- Instructor has identifier Emp ID and has attributes Name, Rank, and Office Phone.
- Timeslot has identifier TSIS and has attributes Day Of Week, Start Time, and End Time.
- Course has identifier Course ID and has attributes Course Description and Credits. Courses can have one, none, or many prerequisites. Courses also have one or more sections.
- Section has identifier Section ID and attribute Enrollment Limit. After some further discussions, you have come up with some additional business rules to help you create the initial design:
 - An instructor teaches one, none, or many sections of a course in a given semester.
 - An instructor specifies preferred time slots.
 - Scheduling data are kept for each semester, uniquely identified by semester and year.
 - A room can be scheduled for one section or no section during one time slot in a given semester of a given year. However, one room can participate in many schedules, one schedule, or no schedules; one time slot can participate in many schedules, one schedule, or no schedules; one section can participate in many schedules, one schedule, or no schedules. Hint: Can you associate this to anything that you have seen before?
- A room can have one type of media, several types of media, or no media.
- Instructors are trained to use one, none, or many types of media.
- A lab has one or more computer types. However, a classroom does not have any computers.
- A room cannot be both a classroom and a lab. There also are no other room types to be incorporated into the system.

Question#3:

A healthcare management system is designed to manage patient information, appointments, medical records, healthcare professionals, and billing processes. The system has the following modules:

- **Patient Registration:**

A Patient registers with the hospital, providing personal information such as name, date of birth, contact details, and insurance information. Each Patient can have multiple visits to the hospital.
- **Appointment Scheduling:**

Patients can schedule Appointments with Healthcare Professionals (Doctors, Nurses). Each Appointment includes a date, time, and purpose of the visit. A Healthcare Professional can have multiple Appointments, and each Appointment can involve multiple Patients in group settings.
- **Healthcare Professionals:**

Healthcare Professionals have various specializations (e.g., General Practitioner, Cardiologist, Surgeon) and are affiliated with specific Departments within the hospital. Each Department can have multiple Healthcare Professionals.
- **Medical Records:**

Each Patient has a Medical Record that includes details of their medical history, medications, allergies, and treatment plans. Each Medical Record can have multiple entries over time, and a Healthcare Professional is responsible for updating it during each visit.

- **Billing and Insurance:**

Each Appointment generates a Bill, which includes charges for services rendered, medications prescribed, and any insurance claims submitted. Patients can have multiple Bills, and each Bill can be associated with one or more Insurance Claims.

- **Pharmacy Management:**

The hospital has a Pharmacy that manages medication inventory. Healthcare Professionals can prescribe medications, which are linked to specific Patients and included in their Medical Records.

- **Lab Tests and Results:**

Patients may undergo Lab Tests (e.g., blood tests, X-rays), which are ordered by Healthcare Professionals. Each Lab Test has associated Results that are recorded in the Patient's Medical Record.

- **Emergency Services:**

The system includes Emergency Services where Patients can be admitted directly. Emergency Cases are recorded separately and can result in immediate treatment plans and urgent billing procedures.

Question# 4:

The administrators of Punjab University want you to design database for their motor vehicle pool. Given that brief summary of operations, draw the appropriate (and fully labelled) ERD. A brief description of operations is as follows:

- Punjab University has many departments.
- Punjab University has many faculty members.
- **A faculty member**, a student can register in **one department at a time**.
- Faculty members may use the vehicles owned by Punjab University for officially sanctioned travel. For example, the vehicles may be used by faculty members to travel to off-campus learning centers, to travel to locations at which research papers are presented, to transport students to officially sanctioned locations, and to travel for public service purposes. The vehicles used for such purposes are managed by Punjab University's Travel Far But Slowly (TFBS) Centre.
- Using reservation forms, each department can reserve **vehicles** for its faculty, who are responsible for filling out the appropriate trip completion form at the end of a trip. **The reservation form** includes the expected departure date, vehicle type required, destination, and name of the authorized faculty member. The faculty member arriving to pick up a vehicle must sign a checkout form to log out the vehicle and pick up a trip completion form. (The TFBS employee who releases the vehicle for use also signs the checkout form.) The faculty member's **trip completion form** includes the faculty member's identification code, the vehicle's identification, the odometer readings at the start and end of the trip, maintenance complaints (if any), gallons of fuel purchased (if any), and the Punjab University credit card number used to pay for the fuel. If fuel is purchased, the credit card receipt must be stapled to the trip completion form. Upon receipt of the faculty trip completion form, the faculty member's department is billed at a mileage rate based on the vehicle type (sedan, station wagon, panel truck, minivan, or minibus) used. (*Hint: Do not use more entities than are necessary. Remember the difference between attributes and entities!*)
- All vehicle maintenance is performed by TFBS. Each time a vehicle requires maintenance, a maintenance log entry is completed on a **pre-numbered maintenance log form**. The maintenance log form includes the vehicle identification, a brief description of the type of maintenance required, the initial log entry date, the date on which the maintenance was completed, and the identification of the mechanic who released the vehicle back into service. (Only mechanics who have an inspection authorization may release the vehicle back into service.)
- As soon as the log form has been initiated, the log form's number is transferred to a **maintenance detail form**; the log form's number is also forwarded to the parts **department manager**, who fills out a **parts usage form** on which the maintenance log number is recorded. The maintenance detail form contains separate lines for each maintenance item performed, for the parts used, and for identification of the mechanic who performed the maintenance item. When all maintenance items have been completed, the maintenance detail form is stapled to the maintenance log form, the maintenance log form's completion

date is filled out, and the **mechanic** who releases the vehicle back into service signs the form. The stapled forms are then filed, to be used later as the source for various maintenance reports.

- TFBS maintains a parts inventory, including oil, oil filters, air filters, and belts of various types. The parts inventory is checked daily to monitor parts usage and to reorder parts that reach the “minimum quantity on hand” level. To track parts usage, the parts manager requires each mechanic to sign out the parts that are used to perform each vehicle’s maintenance; the parts manager records the maintenance log number under which the part is used.
- Each month TFBS issues a set of **reports**. The reports include the mileage driven by vehicle, by department, and by faculty members within a department. In addition, various revenue reports are generated by vehicle and department. A detailed parts usage report is also filed each month. Finally, a vehicle maintenance summary is created each month.